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Improving the robustness of a partial least squares (PLS) model based on pure component selectivity analysis and range optimization: Case study for the analysis of an etching solution containing hydrogen peroxide

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Abstract

Pure component selectivity analysis (PCSA) was successfully utilized to enhance the robustness of a partial least squares (PLS) model by examining the selectivity of a given component to other components. The samples used in this study were composed of NH_4OH, H_2O_2 and H_2O, a popular etchant solution in the electronic industry. Corresponding near-infrared (NIR) spectra (9000–7500 cm^{-1}) were used to build PLS models. The selective determination of H_2O_2 without influences from NH_4OH and H_2O was a key issue since its molecular structure is similar to that of H_2O and NH_4OH also has a hydroxyl functional group. The best spectral ranges for the determination of NH_4OH and H_2O_2 were found with the use of moving window PLS (MW-PLS) and corresponding selectivity was examined by pure component selectivity analysis. The PLS calibration for NH_4OH was free from interferences from the other components due to the presence of its unique NH absorption bands. Since the spectral variation from H_2O_2 was broadly overlapping and much less distinct than that from NH_4OH, the selectivity and prediction performance for the H_2O_2 calibration were sensitively varied depending on the spectral ranges and number of factors used. PCSA, based on the comparison between regression vectors from PLS and the net analyte signal (NAS), was an effective method to prevent over-fitting of the H_2O_2 calibration. A robust H_2O_2 calibration model with minimal interferences from other components was developed. PCSA should be included as a standard method in PLS calibrations where prediction error only is the usual measure of performance.
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Keywords: Near-infrared (NIR) spectroscopy; Pure component selectivity analysis (PCSA); Net analyte signal (NAS); Moving window partial least squares (PLS); Standard clean 1 (SC1) etching solution

1. Introduction

Near-infrared (NIR) spectroscopic monitoring of an etchant solution composition through a Teflon tube has been demonstrated in previous publications [1,2]. For spectral collection, the direct illumination of NIR radiation through an existing Teflon tube is simple and advantageous because all sample lines in a wet station are composed of Teflon tubing [3]. Therefore, an on-line NIR analysis requires minimal system modifications [4–6]. Due to the use of an existing Teflon tube, for example a half-inch tube (inner diameter (pathlength): 9.68 mm), the variation of pathlength for optimal NIR measurement is limited. The 9000–7500 cm^{-1} range is the only choice for the analysis of etchant solutions that contain high concentrations of water. In this range, the broad spectral feature of a combination OH band (centered at 8300 cm^{-1}) is present. Many etchant solutions contain hydrogen peroxide (H_2O_2) as an oxidant and its NIR spectral feature is similar to that of water, especially in the 9000–7500 cm^{-1} range [7]. When partial least squares (PLS) regression is used [8,9], a calibration model should selectively follow the H_2O_2 variation in the presence of hydroxyl (OH)-containing compounds, such as NH_4OH and water. If H_2O_2 variation cannot be selectively described by PLS, the NIR analysis of etchant solutions containing abundant H_2O_2 will not be reliable. Previous publications have demonstrated the practicality of NIR measurement using Teflon tubing. However, calibration selectivity of H_2O_2 has not been investigated extensively. This issue should be addressed to ensure the reliability of NIR mea-

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surement for the analysis of OH-group containing compounds in an aqueous system.

To investigate the calibration selectivity of H_2O_2 in an aqueous environment, an SC1 (standard clean 1) solution [10] composed of NH_4OH, H_2O_2 and H_2O was selected. SC1 solutions are frequently used etchant solutions in electronic industries. The best spectral ranges for the determination of NH_4OH and H_2O_2 were found through the use of moving window PLS (MW-PLS) [11,12]. Corresponding selectivity was examined by pure component selectivity analysis (PCSA) [13]. PCSA is a valuable method for evaluating the calibration selectivity of a given component by comparing the regression vectors from PLS with that from the net analyte signal (NAS) [14,15]. NH_4OH concentration can be measured without any interference from other components due to its unique NH absorption band. Since the OH spectral features of H_2O_2, H_2O and NH_4OH broadly overlapped each other, the calibration selectivity and prediction performance of H_2O_2 were sensitively varied depending on the spectral ranges and number of factors used. Therefore, for robust H_2O_2 calibration, it was essential to optimize the spectral range for PLS and evaluate the calibration selectivity at each case using PCSA.

2. Experimental

2.1. Sample preparation

Forty standard samples were prepared by mixing appropriate amounts of NH_4OH, H_2O_2 and H_2O. The concentration of each component was determined by generating random numbers in a given concentration range to minimize correlations among them. The concentrations of NH_4OH, H_2O_2 and H_2O ranged from 0.56 to 6.41, 2.11 to 8.68 and 86.4 to 96.6 wt%, respectively. The concentration ranges of the standard mixtures were slightly wider than the actual concentration ranges in the process. The chemicals were purchased from common suppliers. Each sample was prepared just prior to NIR spectral collection to minimize analytical error due to evaporation.

2.2. NIR spectral collection and data processing

Fig. 1 shows the configuration used for collecting NIR spectra. A half-inch Teflon tube (outer diameter: 12.7 mm, inner diameter: 9.68 mm) was positioned in the beam path, so the focused NIR beam (approximately 4 mm × 3 mm in size) penetrated the center of the Teflon tube. A Nicolet Magna 550 FT-NIR spectrometer (Therom Nicolet Corp., Madison, WI, USA) equipped with a tungsten–halogen source and cryogenically cooled InSb was used. All NIR spectra corresponded to 128 scans with a resolution of 32 cm^{-1}. NIR spectra were collected at room temperature immediately after sample preparation. A total of 120 spectra (triplicate spectra for each sample) from 40 samples were divided into 96 spectra (32 samples) for calibration and 24 spectra (8 samples) for validation. Spectra for the validation set were randomly chosen. All spectral processings including partial least squares regression, principal component analysis (PCA) [16] and pure component selectivity analysis were done using Matlab Version 7.0 (MathWorks Inc., MA, USA).

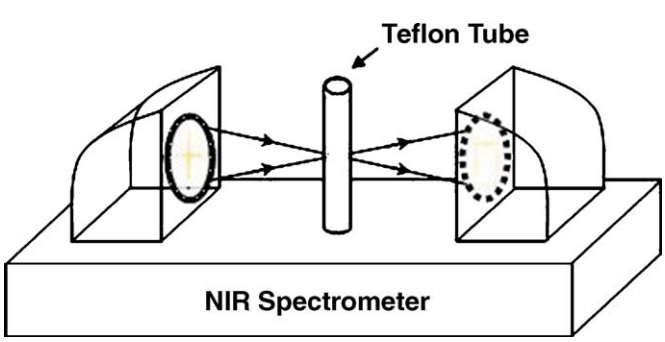

Fig. 1. Schematic diagram of the instrumental configuration for spectral collection using Teflon tubing.

3. Results and discussion

3.1. NIR spectral features of each component

The spectral feature of water was examined first since it was the solvent for the SC1 solution. Fig. 2(a) shows the NIR water spectrum (bold line) in the 9000–7500 cm^{-1} range. The

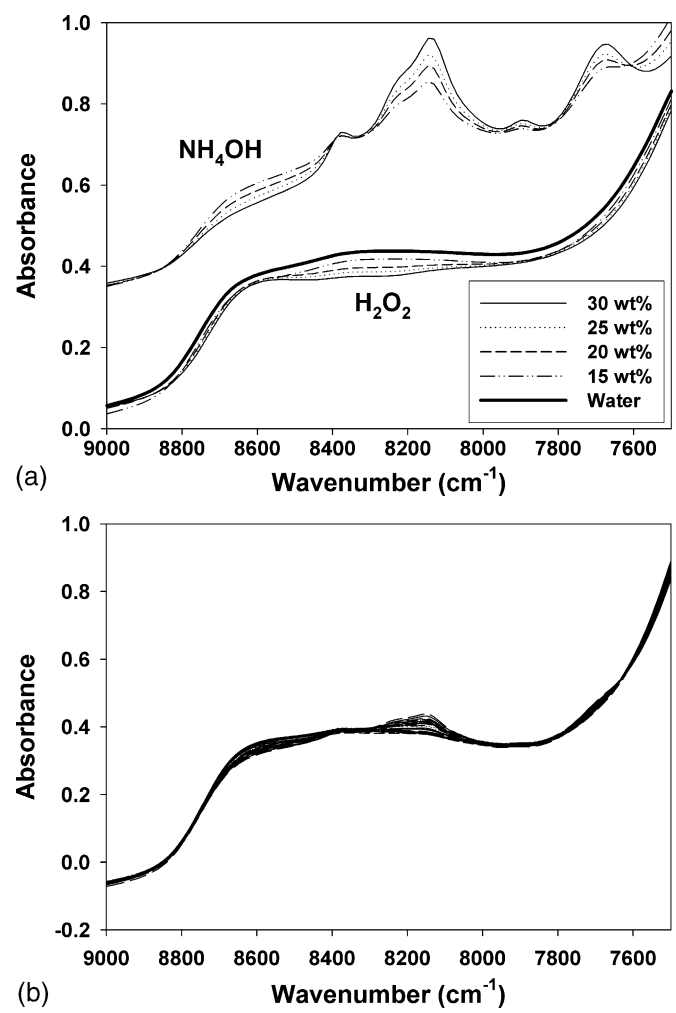

Fig. 2. NIR spectra (9000–7500 cm^{-1} range) of water and 15, 20, 25 and 30 wt% NH_4OH and H_2O_2 (a) and 40 different mixture samples (b).

broad water absorption band was observed in the range of 8800–7800 cm^{-1} with a maximum absorption at 8300 cm^{-1}. The water band corresponds to the combination band of OH (e.g. ≈6700 cm^{-1} + 1600 cm^{-1} (bending) = 8300 cm^{-1}). The samples used in this study contained high concentrations of water, a strong NIR absorber. Therefore, the spectral features of the SC1 solutions will be the spectral variation of each component overlayed with the water band.

To investigate the spectral variations of each component, simple binary mixtures of NH_4OH–water and H_2O_2–water were prepared. The corresponding NIR spectra were examined. Fig. 2(a) shows the NIR spectra of 15, 20, 25 and 30 wt% NH_4OH and H_2O_2. The spectra of NH_4OH were offset for a better visual comparison. With increasing NH_4OH concentrations, distinct and systematic spectral variations were observed at 8600, 8180 and 7690 cm^{-1}. The 8180 and 7690 cm^{-1} bands correspond to the combination bands of NH vibration. Even though two NH bands appeared on the broad water band, these bands become more dominant with increasing NH_4OH concentration due to distinct differences in the molecular structure of NH_4OH. The OH band in the 8800–8400 cm^{-1} range weakened due to water displacement from the relative decrease of water concentration.

The spectral variation of H_2O_2 is different from that of NH_4OH. Spectral changes originating from the variation of H_2O_2 appears in the 8600–7900 cm^{-1} range. The broad spectral variation occurred due to the similarity of molecular structures. The only observable feature is the decrease of the OH band in the 8600–7900 cm^{-1} range with the increase of H_2O_2 concentration. This trend is due to water displacement by H_2O_2.

Fig. 2(b) shows 40 different mixture spectra of NH_4OH, H_2O_2 and H_2O used in this study. The most significant variations were observed at 8180 and 7690 cm^{-1}. This is where the NH_4OH spectral features are located and the magnitude of spectral variation of NH_4OH is larger than that of H_2O_2. Since the spectral changes from H_2O_2 variation are broad and relatively small, they are difficult to identify visually. The spectral variation originating from H_2O_2 is not characteristic. However, the spectral variation from NH_4OH is distinct. Consequently, logical and optimal calibration strategies will be necessary to selectively follow H_2O_2 variation by minimizing interference from other components (NH_4OH and H_2O) that contain a hydroxyl group.

3.2. PLS calibration and pure component selectivity analysis in the whole spectral range

Partial least squares regression was used to build calibration models for each component. As in previous publications, PLS was initially performed using the whole spectral range (9000–7500 cm^{-1}) where all of the spectral features were available. The optimum number of factors was identified by examining the decreasing pattern of standard error of calibration (SEC) as a function of the number of PLS factors. The standard error of prediction (SEP) was calculated by predicting the samples in the validation set using a developed model.

Pure component selectivity analysis was performed to verify the selectivity of the PLS model. For this purpose, the regression vectors from the PLS and net analyte signal were generated and corresponding features were compared as in a previous publication [1,2]. This comparison can indicate how selectively a developed PLS model describes the variation of a target analyte in a mixture system. When the pattern of regression vectors from PLS and NAS from a pure component is similar, the developed PLS model selectively describes the variation of a target analyte with minimal or no interferences from other components. To calculate the regression vectors from NAS for NH_4OH and H_2O_2, individual pure component spectra in the 15–30 wt% range were acquired in the 9000–7500 cm^{-1} range.

The overall calibration results for NH_4OH are presented in Fig. 3 and Table 1. Fig. 3(a) shows the SEC (open circles) and SEP (filled circles) plotted as a function of the number of PLS factors used to determine NH_4OH in the 9000–7500 cm^{-1} region. The SEC sharply decreases for the first factor and gradually decreases when more analyte-related spectral variation is incorporated into the calibration model. However, no significant improvement in SEC was observed after two PLS factors. Thus, two factors were chosen as the optimum. The resulting SEP was lowest with two factors. The numbers in parentheses in Table 1

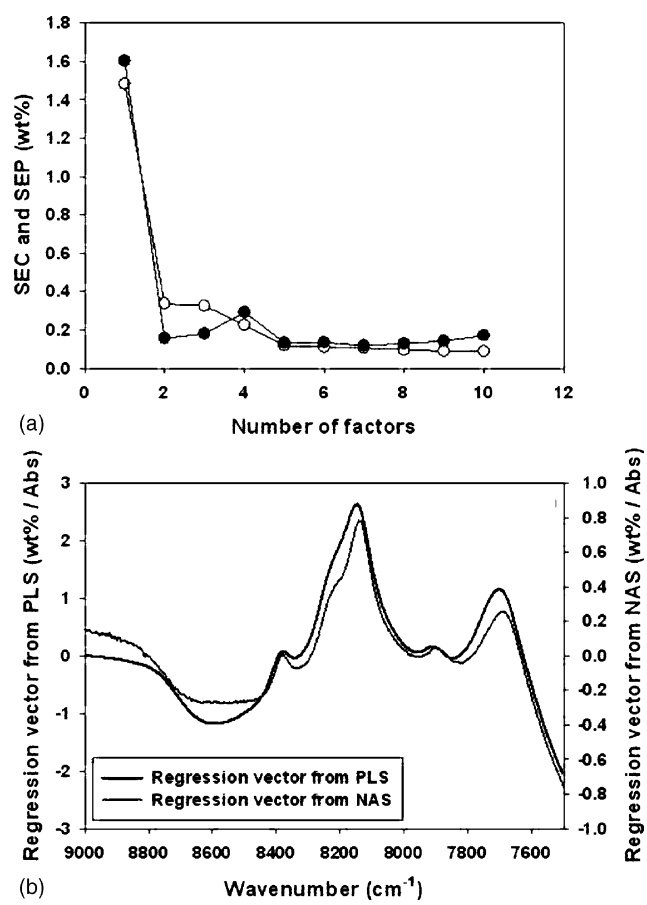

Fig. 3. The SEC (open circles) and SEP (filled circles) plotted as a function of the number of PLS factors used to determine NH_4OH in the 9000–7500 cm^{-1} region (a) and the corresponding NH_4OH regression vectors from PLS and NAS in the same region (b).

Table 1
PLS calibration results for NH_4OH and H_2O_2 using the whole spectral range (9000–7500 cm^{-1})

Spectral range (cm^{-1})	NH_4OH		H_2O_2	
	SEC (wt%)	SEP (wt%)	SEC (wt%)	SEP (wt%)
9000–7500	0.33 (2)	0.15	0.26 (5)	0.20
9000–7500	–	–	0.09 (6)	0.13

The numbers in parentheses correspond to the number of factors used.

corresponds to the numbers of PLS factors used. The achieved SEC and SEP were 0.33 and 0.15 wt%, respectively. The SEC is better to be improved.

Fig. 3(b) shows regression vectors from PLS and NAS in the 9000–7500 cm^{-1} region. The overall shapes of the two regression vectors were similar and the major bands of NH_4OH at 8180 and 7690 cm^{-1} were clearly identified. As mentioned, the similarity of the two vectors from PLS and NAS is an indication of good selectivity of a developed PLS model that leads to an overall robustness of calibration.

Fig. 4 shows the residual plots from both calibration and prediction as a function of H_2O_2 (top) and water (bottom) concentration variations. Open and filled circles correspond to calibration and prediction data, respectively. These plots were designed to examine the existence of possible prediction dependency by other components. There is no systematic trend in prediction, such as slope and the prediction data are evenly scattered without influences from the variations of the other two components. Overall, it can be concluded that the developed PLS model for NH_4OH using the whole spectral range is proper and selective to other components.

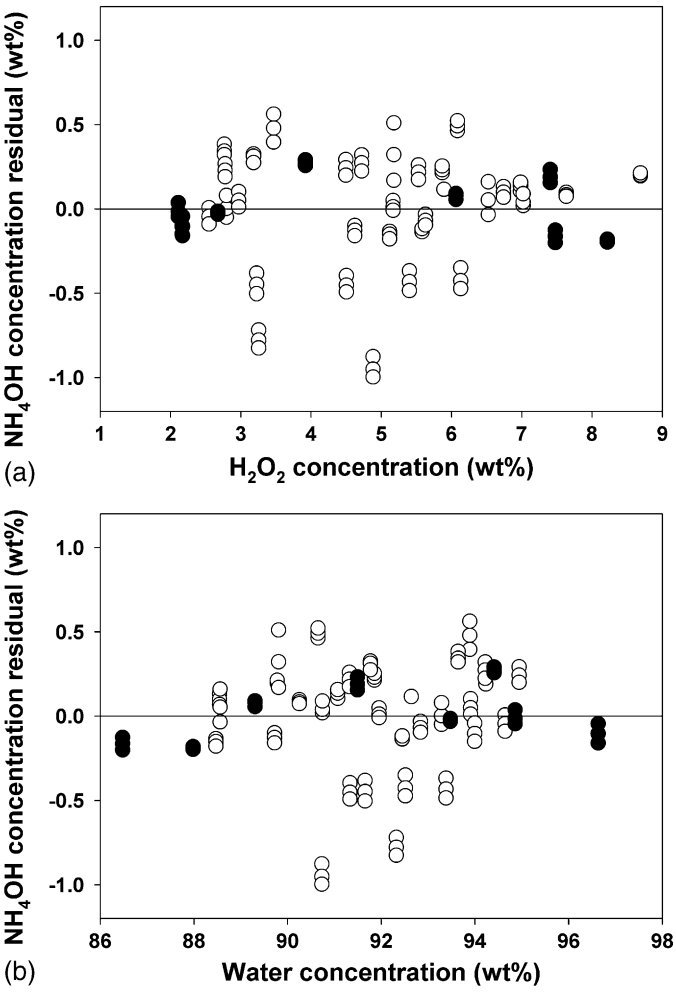

Fig. 4. NH_4OH residual plots from calibration (for the 9000–7500 cm^{-1} range with two factors) and prediction as a function of H_2O_2 (top) and water (bottom) concentration variations. Open and filled circles correspond to calibration and prediction data, respectively.

Fig. 5 shows the SEC (open circles) and SEP (filled circles) plotted as a function of the number of PLS factors used to determine H_2O_2 in the 9000–7500 cm^{-1} region. The pattern is different from that of NH_4OH where a sharp drop in SEC was observed at the initial factor. The SEC decrease pattern of H_2O_2 is slight since its spectral feature is broad and less distinct compared to NH_4OH. Also, it required more factors to follow the variation of H_2O_2. Based on Fig. 5, five or six PLS factors could be chosen. Therefore, PCSA was performed for both cases and corresponding results were evaluated.

Fig. 6 shows the H_2O_2 regression vectors from PLS (five factors) and NAS in the 9000–7500 cm^{-1} region. It also shows residual plots from the corresponding calibration and prediction as a function of NH_4OH and water concentration variations. The resulting SEC and SEP are also summarized in Table 1. When five PLS factors were used, the overall shapes of the regression vectors from PLS and NAS of H_2O_2 were similar. However, a slight discrepancy was observed in the 8200–7500 cm^{-1} range. The magnitude of the absorption changes based on H_2O_2 is small and its spectral features are broad. Therefore, it is difficult

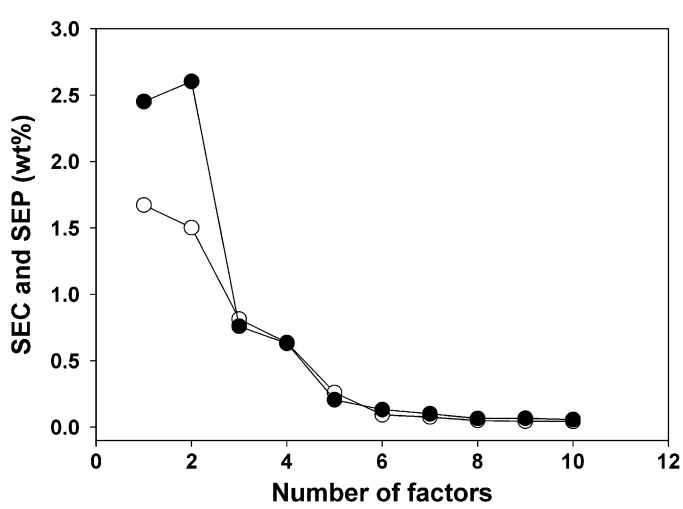

Fig. 5. SEC (open circles) and SEP (filled circles) plotted as a function of the number of PLS factors used to determine H_2O_2 in the 9000–7500 cm^{-1} region.

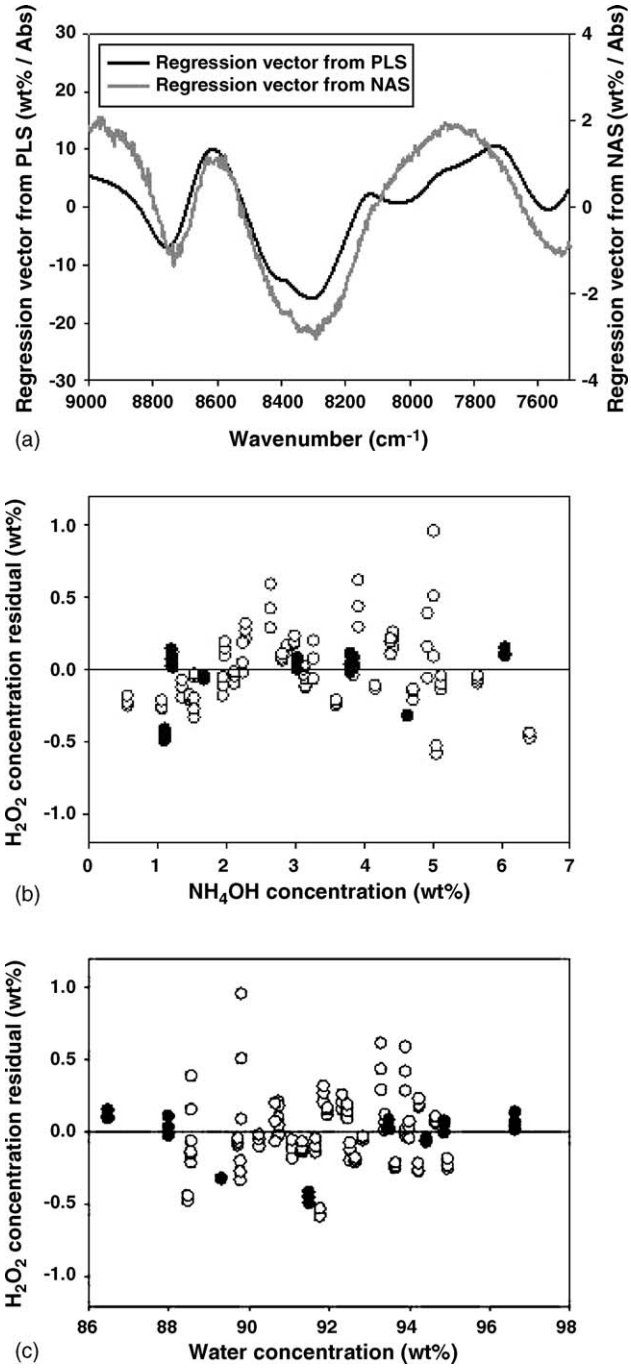

Fig. 6. H_2O_2 regression vectors from PLS (five factors) and NAS in the 9000–7500 cm^{-1} region (a), the residual plots from the corresponding calibration and prediction as a function of NH_4OH (b) and water (c) concentration variations.

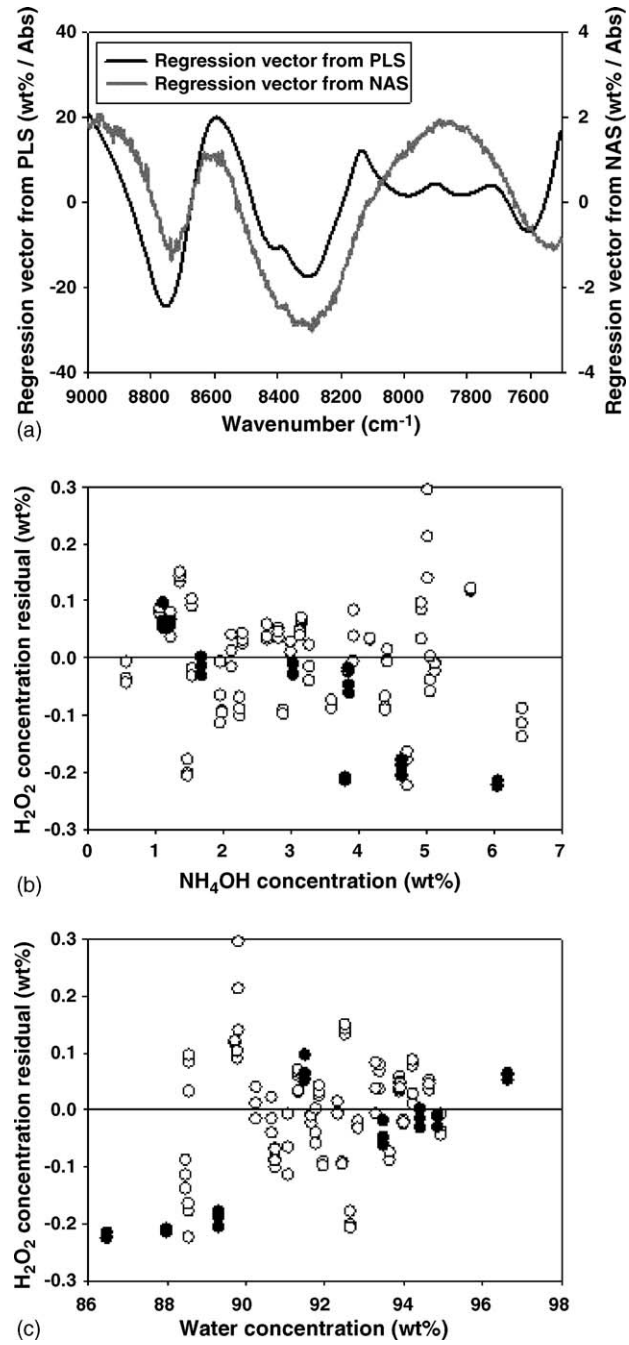

Fig. 7. H_2O_2 regression vectors from PLS (six factors) and NAS in the 9000–7500 cm^{-1} region (a), the residual plots from the corresponding calibration and prediction as a function of NH_4OH (b) and water (c) concentration variations.

to follow only the H_2O_2 variation compared to NH_4OH. In the residual plots, no systematic prediction pattern is observed and the prediction data are evenly scattered as with NH_4OH. The five factor model describes the variation of H_2O_2 selectively, but the resulting SEP is somewhat high (0.20 wt%).

As shown in Table 1, SEP could be improved to 0.13 wt% if six PLS factors are used to build a model. However, this may not be a true improvement to selectively describing the H_2O_2 variation. Fig. 7 shows the same plots as Fig. 6 except the results are

Table 2
PLS calibration results for NH_4OH and H_2O_2 using the optimized spectral ranges for each component

Spectral range (cm^{-1})	NH_4OH		H_2O_2	
	SEC (wt%)	SEP (wt%)	SEC (wt%)	SEP (wt%)
8050–7550	0.13 (3)	0.13	–	–
8850–8050	–	–	0.13 (4)	0.16
8675–8050	–	–	0.09 (5)	0.12

The numbers in parentheses correspond to the number of factors used.

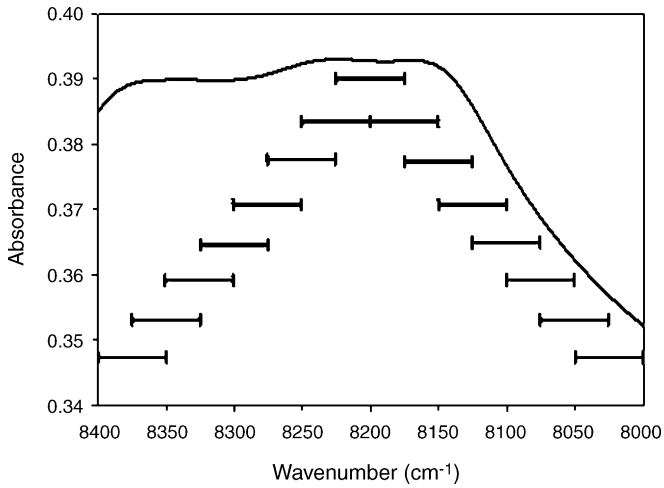

Fig. 8. Simplified scheme showing the procedure of moving window PLS (MW-PLS). The bars correspond to window size (50 cm^{-1} in this case) and move from right to left in 25 cm^{-1} steps.

from using six PLS factors. A large discrepancy in the regression vectors from PLS and NAS was observed when compared to Fig. 6(a). The difference between both regression vectors was increased in the 9000–8300 cm^{-1} range (rich hydroxyl information range). The regression vector from the PLS appeared to add NH_4OH information in the 8300–7600 cm^{-1} range since its shape is somewhat similar to that in Fig. 3(b) and the characteristic NH band at 8180 cm^{-1} was observed. The residual plots clearly demonstrate the H_2O_2 prediction dependency on other components (NH_4OH and water). Slopes are clearly observed in both cases and the directions of both slopes are opposite. These results show that the six factor model fails to describe H_2O_2 variation selectively even though the resulting SEP is decreased. The SEP was improved by incorporating partial information from NH_4OH and water. This is a typical example of over-modeling of PLS that should be avoided. PCSA (comparison of regression vectors from PLS and NAS) is an effective way to examine the selectivity of calibration without over-fitting, especially when spectral features are broad and overlapping.

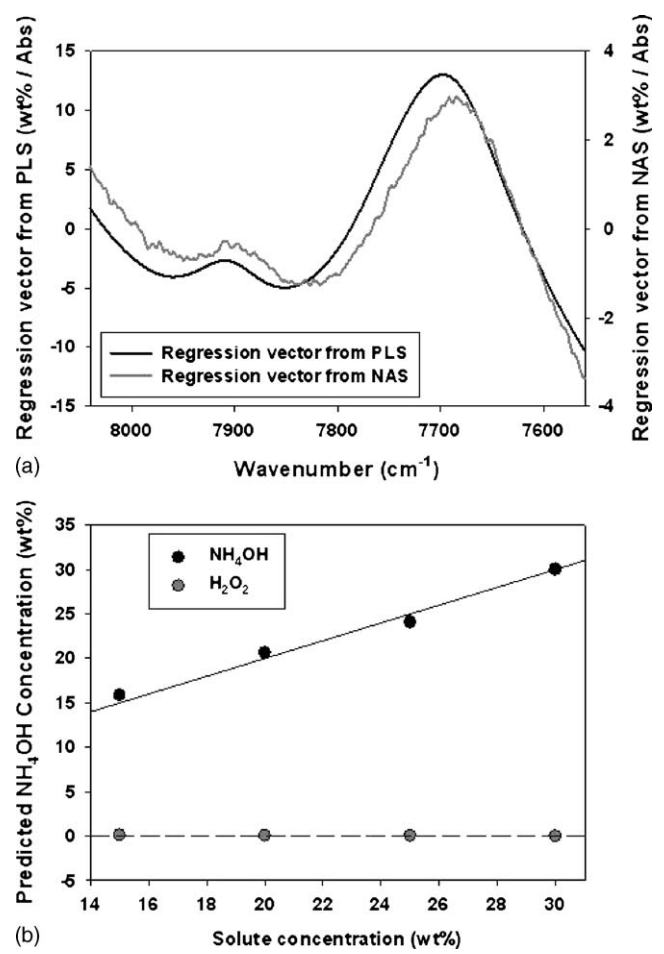

Fig. 9. NH_4OH regression vectors from PLS and NAS in the 8050–7550 cm^{-1} range (a) and the selectivity plot of NH_4OH over H_2O_2 (b).

3.3. Range optimization for PLS and corresponding pure component selectivity analysis

The use of the whole spectral range for PLS could be advantageous since all the information can be utilized. However, this

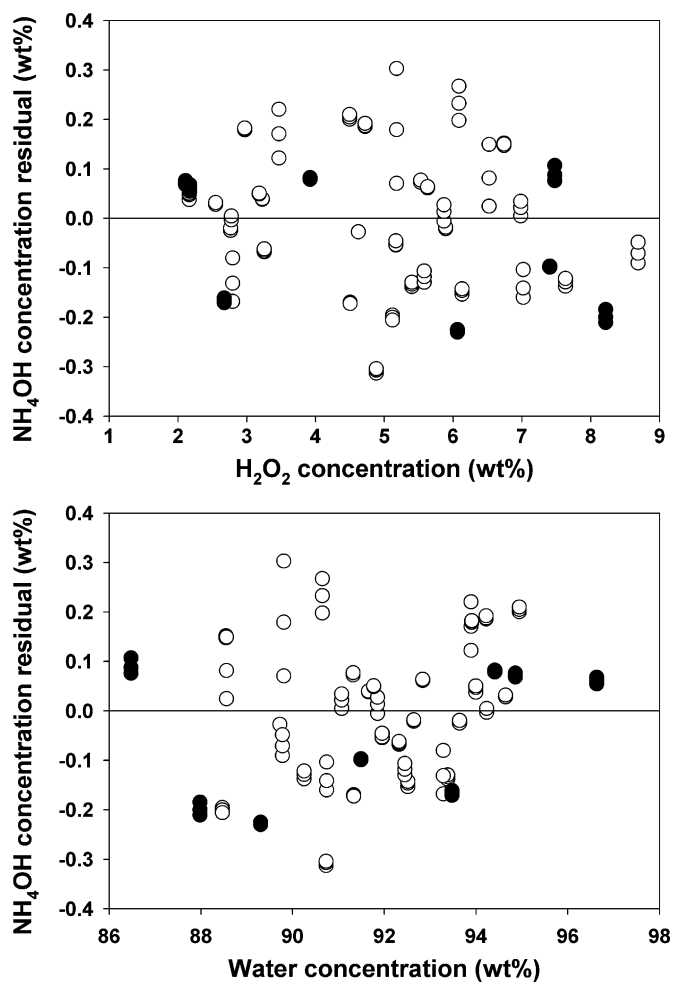

Fig. 10. NH_4OH residual plots from calibration (for the 8050–7550 cm^{-1} range with three factors) and prediction as a function of H_2O_2 (top) and water (bottom) concentration variations. Open and filled circles correspond to calibration and prediction data, respectively.

might not always be accurate when the spectral features of a given analyte broadly overlap with those from other components. Possibly, an optimal spectral range exists that yields improved calibration results. Several chemometric strategies, such as moving window PLS, for finding the best calibration range have been reported. A similar method of MW-PLS was used in this study. The purpose was to perform PLS at all possible combinations of spectral ranges and find the spectral range that yields a minimal error. The window size (number of spectral points) was varied from 50 to 1500 cm^{-1} at increments of 50 cm^{-1}. In a given window size, the window was moved from low to high wavenumbers in 25 cm^{-1} steps until the whole spectral range (9000–7500 cm^{-1}) was covered. This procedure is shown in Fig. 8. Only the 8400–8000 cm^{-1} range is shown to demonstrate the concept of MW-PLS. The bars correspond to window size (50 cm^{-1} in this case) and move from right to left in 25 cm^{-1} steps. To perform MW-PLS, the original calibration set (96 spectra from 32 samples) was divided into 78 spectra (26 samples) for a new calibration set and 12 spectra (6 samples) for a new validation set. At each spectral range (bar), PLS was performed using the new calibration set and the resulting SEP from the new validation set was evaluated. These new calibration and validation sets were used only for MW-PLS. A total of 841 spectral ranges with different window sizes and different spectral ranges were tested.

For NH_4OH, the best results were acquired using three factors for the 8050–7550 cm^{-1} range, where the 7690 cm^{-1} NH band was highlighted. The results are summarized in Table 2. The other NH band at 8180 cm^{-1} provides enough quantitative information, but it overlaps with the OH band, as shown in Fig. 1(a). Therefore, the use of the less-influenced 7690 cm^{-1} NH band was preferred.

PCSA was performed using the optimized range (8050–7550 cm^{-1}) and the results are shown in Fig. 9. The regression vectors from PLS and NAS are similar to each other and the 7690 cm^{-1} NH band is easily identified. The PLS regression vector accurately describes the NAS of NH_4OH in the 8050–7550 cm^{-1} range. Fig. 9(b) shows the selectivity plot of NH_4OH over H_2O_2. The concentration of NH_4OH was calculated by multiplying the NH_4OH NAS regression vector and the absorbance from pure NH_4OH. The concentration of H_2O_2 was calculated by multiplying the NH_4OH NAS regression vector and the absorbance from pure H_2O_2. If the selectivity of NH_4OH is not satisfactory, the calculated NH_4OH concentration will deviate from the ideal line (solid line) and the calculated H_2O_2 concentration will depart from the zero-residual line (dashed line). Based on Fig. 9(b), NH_4OH

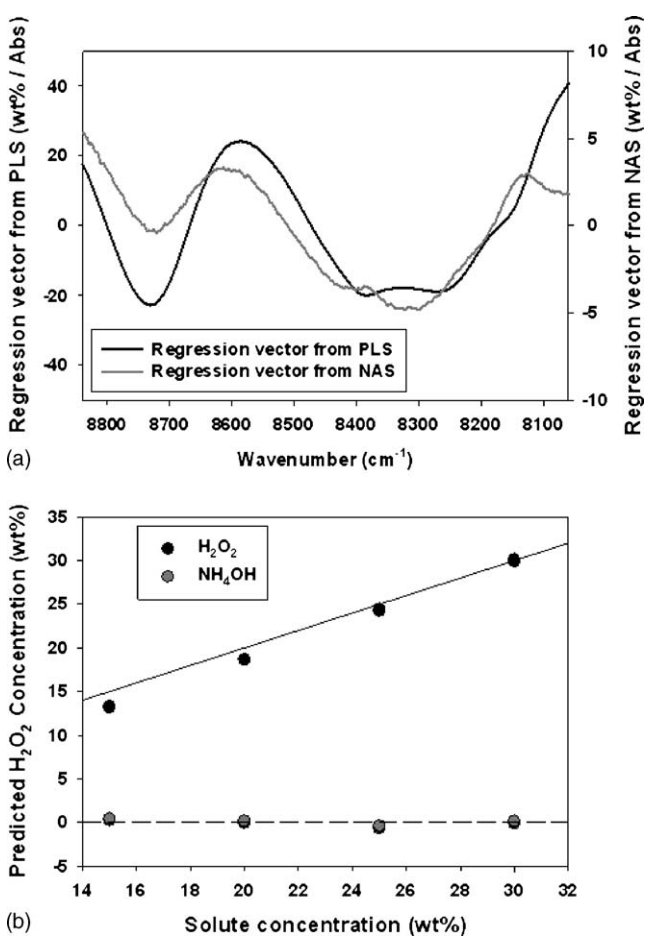

Fig. 11. H_2O_2 regression vectors from PLS and NAS in the 8850–8050 cm^{-1} range (a) and the selectivity plot of H_2O_2 over NH_4OH (b).

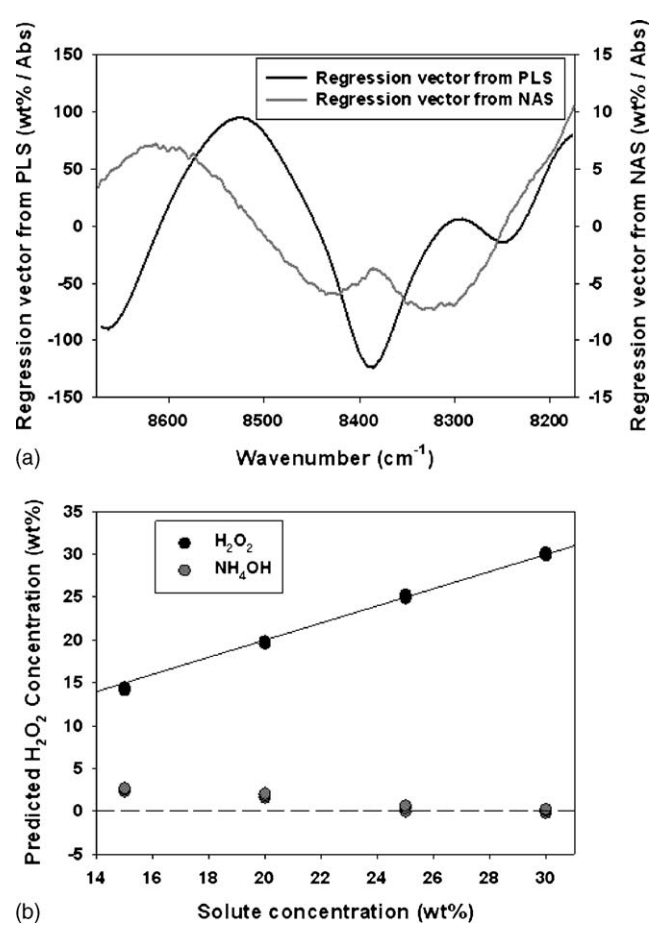

Fig. 12. H_2O_2 regression vectors from PLS and NAS in the 8675–8175 cm^{-1} range (a) and the selectivity plot of H_2O_2 over NH_4OH (b).

concentration is effectively described by the NAS regression vector without influence from H_2O_2 variations.

Fig. 10 shows the corresponding residual plots from both calibration and prediction as a function of H_2O_2 (top) and water (bottom) concentration variations. Open and filled circles correspond to calibration and prediction data, respectively. There was no systematic trend in prediction and the prediction data are evenly scattered. The optimal calibration model for NH_4OH was acquired using the 8050–7550 cm^{-1} range with three factors. The resulting SEC and SEP were improved to 0.13 wt%. The model selectively followed the NH_4OH variation without any prediction slope or bias.

For H_2O_2, the best results were acquired at two different spectral ranges: the 8850–8050 cm^{-1} range using four factors and the 8675–8050 cm^{-1} range using five factors. The resulting SEC and SEP are summarized in Table 2. For both spectral ranges, the regression vectors from PLS and NAS were compared and selectivity plots were examined. The resulting plots are shown in Fig. 11 (8850–8050 cm^{-1} range) and Fig. 12 (8675–8175 cm^{-1} range). The regression vectors from PLS and NAS are better matched in the 8850–8050 cm^{-1} range while the SEP in the 8675–8175 cm^{-1} range is superior. The selectivity plots of Fig. 11(b) and Fig. 12(b) generated from both spectral ranges are slightly different each other. For the 8850–8050 cm^{-1} range, the H_2O_2 selectivity over NH_4OH is good. However, the description of H_2O_2 slightly deviated from the ideal line (solid line). Conversely, in the 8675–8175 cm^{-1} range, the description of H_2O_2 follows the ideal line fairly well, but the H_2O_2 selectivity over NH_4OH was degraded. Neither of the two PLS models are perfect, but the model developed in the 8850–8050 cm^{-1} range using four PLS factors is superior because the regression vectors from PLS and NAS are better matched.

Figs. 13 and 14 show residual plots as functions of the variation in NH_4OH and water concentration when the 8850–8050 cm^{-1} and 8675–8175 cm^{-1} range were used, respectively. Open and filled circles correspond to calibration and prediction results, respectively. In Fig. 13, no systematic trend was observed in the plots using the 8850–8050 cm^{-1} range. However, slopes were observed in both residual plots in Fig. 14.

Based on the selectivity plots of H_2O_2, it was difficult to follow the variations of H_2O_2 without influences from NH_4OH and

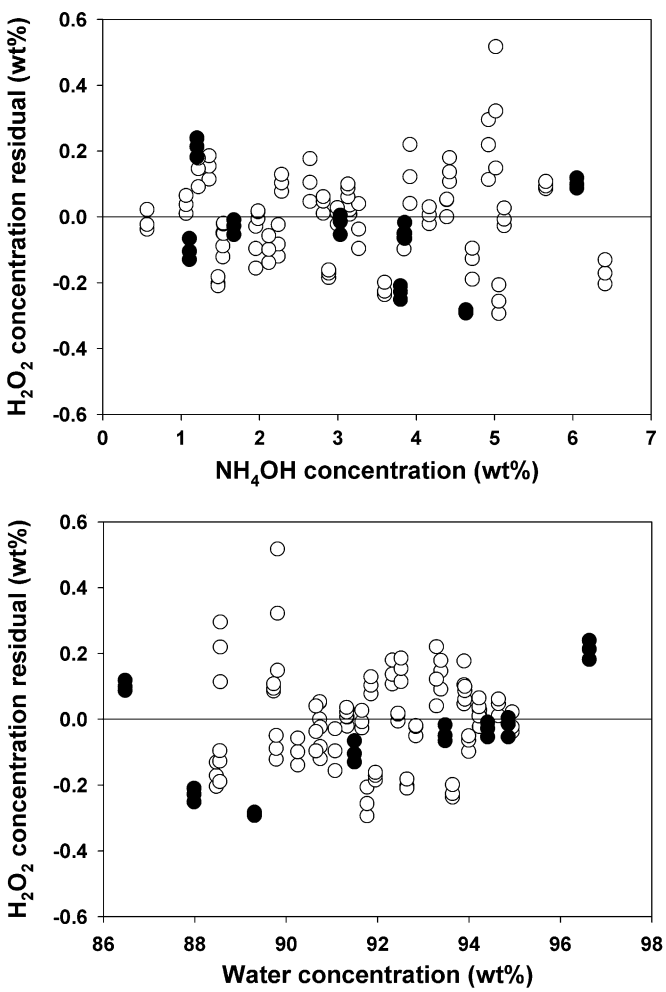

Fig. 13. H_2O_2 residual plots as a function of NH_4OH (a) and water (b) concentration variations when the 8850–8050 cm^{-1} range with four factors was used. Open and filled circles correspond to calibration and prediction data, respectively.

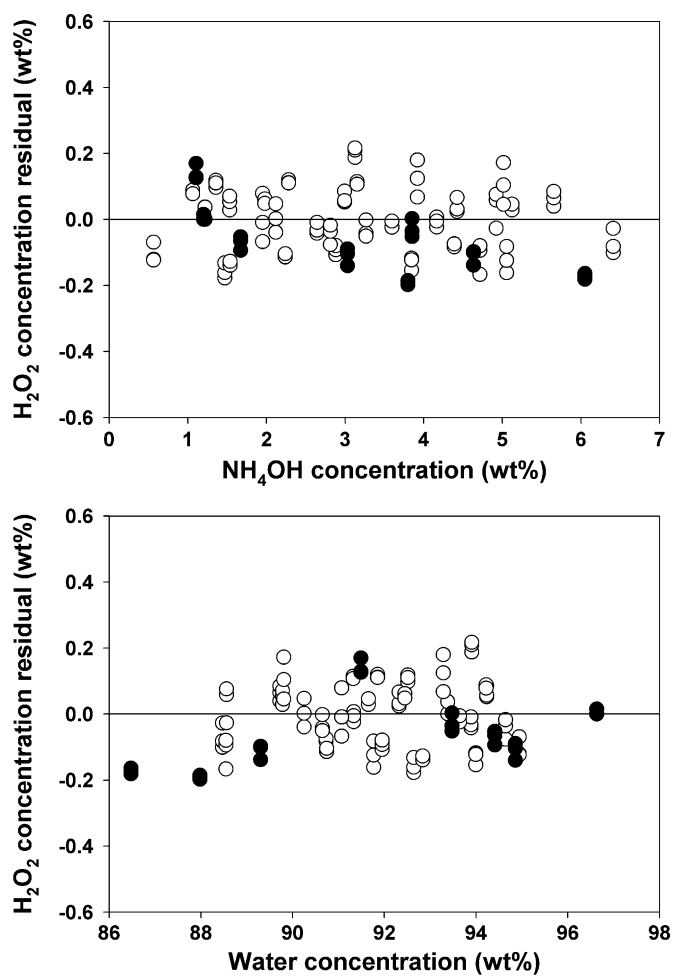

Fig. 14. H_2O_2 residual plots as a function of NH_4OH (a) and water (b) concentration variations when the 8675–8175 cm^{-1} range with five factors was used. Open and filled circles correspond to calibration and prediction data, respectively.

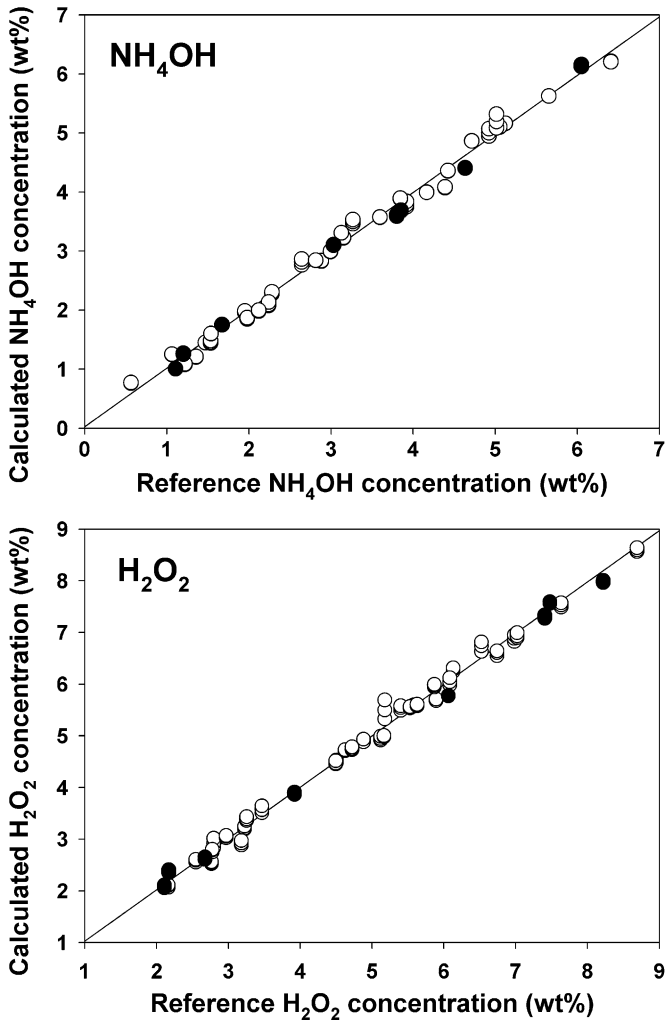

Fig. 15. Scatter plots show the correlation between reference concentrations and NIR measurements for NH_4OH and H_2O_2 based on the best PLS models for each component. Open and filled circles represent the calibration and prediction data, respectively.

water. However, by balancing selectivity and accuracy, an optimal PLS model was developed using the 8850–8050 cm^{-1} range and four PLS factors. The resulting SEC and SEP improved to 0.13 and 0.16 wt%, respectively, compared to the use of whole spectral ranges.

The scatter plots in Fig. 15 show the correlation between reference concentrations and NIR measurements for NH_4OH and H_2O_2 based on the best PLS models for each component (NH_4OH: 8050–7550 cm^{-1} range using three factors, H_2O_2: 8850–8050 cm^{-1} range using four factors). Open and filled circles represent the calibration and prediction data, respectively. The calibration and prediction data correlate well with the reference data for both cases. With the help of PCSA, robust and selective PLS models were developed. These models were particularly useful and necessary for H_2O_2 determination.

4. Conclusion

The importance of selectivity analysis in PLS calibrations using real industrial samples was demonstrated. For robust calibration, the optimal spectral range should be searched and the corresponding selectivity and accuracy of the model must be considered using PCSA. A simple PLS calibration using a wide spectral range without concern for the fundamentals of multivariate calibration can lead to misleading prediction results. This is especially true for the quantitative analysis of OH-containing compounds in an aqueous system, such as an SC1 solution. In the future, a similar analytical approach will be performed for a naphtha sample, a complex mixture of C_6–C_9 paraffins, isoparaffin, naphthene (cycloalkane) and aromatic hydrocarbons [17]. The spectral features of naphtha are highly overlapping due to similarities in the molecular structures and its compositional complexity. Even though NIR spectroscopy has been used extensively for compositional analyses of naphtha, the selectivity of a PLS model for a specific component should be used to confirm the validity of the NIR analysis.

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